KEXIN FU

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Research Profile

Third-year PhD candidate in Biostatistics concentration with an MS degree in Biostatistics. Experienced in determining research estimand, developing missing-data methods, designing and conducting simulation study and sensitivity analysis, analyzing high-dimensional biomarkers and survival outcomes with real-world clinical data. Build reproducible workflows in R and collaborate with clinicians on statistical analysis planning and power calculations.

Education Background

George Washington University, Milken Institute School of Public Health *08/2024-05/2029 (Expected)*

PhD in Biomedical and Health Data Science, Biostatistics concentration (GPA:3.95/4.0)

- Third academic year expected to be completed by May 2027

New York University, School of Global Public Health *09/2022-07/2024*

MS in Biostatistics (GPA:3.9/4.0), Advanced certificate of Public Health Data Science

Relevant coursework: Statistical inference, Statistical computing, Principle/Advanced clinical trial, Longitudinal analysis, Causal inference, Survival analysis, Machine learning, Neural network, High performance and cloud computing

Technical Skills

Programming: R — simulations, applied analysis, visualization, and Shiny workflows; SAS — clinical-trial data extraction and filtering; Python — machine-learning and deep-learning projects

Computing: High-performance computing and AWS EC2

Methods: Missing data imputation, large-scale parallel simulation, high-dimensional biomarker analysis, survival analysis, longitudinal analysis, power analysis, reproducible workflow, multidisciplinary collaboration

Work experience

Biostatistician, GWU Biostatistics Center *10/2024-Present*

- Missing data project (Funded by *GWSPH Research Innovation Award)*

- Develop an extension of multiple imputation by chained equations to jointly impute scalar and functional measurements using functional principal components and Rubin's rules.
- Design and implement parallel simulations in R on high-performance computing to evaluate bias, empirical coverage, and efficiency across data-generating and missingness scenarios.
- Apply and validate the workflow using clinical and EEG data, including convergence diagnostics; the project received first place in Biostatistics & Bioinformatics at GWSPH Research Day 2026.

- Biomarker projects, Diabetes Prevention Program Outcomes Study (DPPOS)

- Co-lead statistical analysis plan development and power analyses for longitudinal proteomic biomarker studies, collaborating with clinicians on quality control and downstream analyses.
- Develop quality-control procedures for extracellular vesicle biomarkers and assess their impact on analyses of insulin signaling and associations with Alzheimer's disease and related dementias.
- Build reproducible R and Shiny workflows for biomarker portfolios.

Biostatistical Consultant, NYU Biostatistical Collaboration and Consultation Core *01/2024-11/2025*

- Collaborated with NYU Langone and external organizations to co-lead the statistical analyses as primary biostatistician.
- Rare disease survival project: Conducted time-to-event analyses using composite endpoints and sensitivity analyses to characterize survival and clinical progression in a real-world Barth syndrome cohort
- Longitudinal tobacco project: Built the longitudinal cross-lagged model studying e-cigarette exposure effect on long-term mental health disorders using longitudinal PATH data.

GPH Course Assistant & Graduate Student Mentor, NYU *06/2023-05/2024*

- Supported 4 graduate-level courses: Applied Survival Analysis, Linear Regression and Modeling, Data Management and Statistical Computing, and Biostatistics for Public Health.
- Guided undergraduates to perform and present aging-related analysis in PQAR 2023 summer program as mentor.

Honors/Awards

04/2026 1st place for Research staff/Postdoc/Doctroral/Alumini in the Biostatistics and Bioinformatics area of GWSPH Research Day: *Functional Regression-based Multiple Imputation for Mixed Missing Data*

03/2026 ***ENAR DataFest 2026 Winner (GW team leader): Integrating Multi-Omics Data for Breast Cancer Survival Prediction Using Machine Learning Methods***

05/2024 NYU Thesis Recognition Award - Innovative Topic or Approach: Neuropathological Measurements and Progression Rate of Alzheimer's Disease: Addressing Racial Disparities and Optimizing Trial Efficiency

05/2024 NYU GPH Biostatistics Community Recognition Award

12/2019 National Scholarship of China

Publications

• **K.Fu,** Y.Huang, V.Bowen, et al., "Survival and Clinical Progression in Barth Syndrome: Insights From the Barth Syndrome Foundation's Database of 502 Affected Individuals," Journal of Inherited Metabolic Disease49, no. 3 (2026): e70189, https://doi.org/10.1002/jimd.70189

• Zhao, H., Zhou, X., **Fu, K**. et al. Prospective cohort study evaluating the association between influenza vaccination and neurodegenerative diseases. npj Vaccines 9, 51 (2024). https://doi.org/10.1038/s41541-024-00841-z

• Lei Wan, Fangzhou Geng, **Kexin Fu**, Wei chen. Low-intensity pulsed ultrasound in postoperative fracture healing: Current research landscape and hotspot analysis. Asian Journal of Surgery, 2025, ISSN 1015-9584. https://doi.org/10.1016/j.asjsur.2025.06.198

Contributed Talks/Posters

• Adam Ciarleglio [P,1], **Kexin Fu[1],** *Functional principal components-based multiple imputation for functional data* **(JSM 2026 Conference contributed paper)**

• **Kexin Fu** [P,1], Yajie Li [P,1], Qixiang Chen [P,1], Josh T'Korley [P,2], *Integrating Multi-Omics Data for Breast Cancer Survival Prediction Using Machine Learning Methods* ***(ENAR 2026 DataFest invited presentation)***

• *The Prevention and Treatment of Missing Data in Regulatory Clinical Trials* (*PUBH 6866* ***Principle of clinical trials*** **Special Topic Invited Lecturer** at GWSPH)

• **Kexin Fu** [P,1],Yan Luo [2], *Comparison of Surgical Outcomes Between Two Types of cervical polypectomy during pregnancy* **(ENAR 2025 Conference poster)**

• **Kexin Fu** [P,1], Han Oo [1], Carolyn Taylor [2], Rachel Monahan [3], Sang Lee [3], Thuy Linh Nguyen [4], Thao Vu [5], Mai T. Do [5], Thanh Huong T. Tran [5], PhuongThao D. Le [6], *Evaluating Feasibility and Adaptability of Stronger Together Peer Mentoring Program for Cancer Patients in Viet Nam Using the Consolidated Framework for Implementation Research (IPOS 2024 Conference poster)*

• **Kexin Fu** [P,1], Jingyuan Chen [2], Manali Ashok Dule [2], Alex Dahlen [2], Mattia Gilmartin [3], *Evaluating the NICHE Program: Disparities in Aging Care Implementation Across US Hospitals* (Howard-Meyers Bernadine Lacey Health Equity 2024 Conference poster)

Volunteers

09/2026 ASA Biopharmaceutical Section RISW Volunteer

06/2026 ICSA Volunteer

02/2026-Present **PhD Student Representative**, GWU School of Public Health

09/2025 ASA Biopharmaceutical Section RISW Volunteer

Professional Development/Memberships

• **Advanced Estimands and Missing Data Imputation in Clinical Trials** — ASA RISW Short Course, 2026

• **Graphical Multiple Comparison Procedures: Combining Flexibility with Optimality** — ASA RISW Short Course, 2025

• American Statistical Association — Student Member

• Washington Statistical Society —— Student Member

• International Biometric Society, Eastern North American Region — Student Member

• International Chinese Statistical Association —— Student Member